Tutorato di algoritmi e laboratorio 6

Piccola digressione ...

perchè dico spesso "di conseguenza?"

Modus Ponens $$\frac{\alpha \qquad \alpha \to \beta}{\beta}$$

Esempio: "So che la mia stima della distanza tra i vertici a e c è 11. So che andare da a in b mi costa 5 e andare da b in c mi costa al più 2.
So che $5+2=7<11$.
Di conseguenza mi conviene stimare la distanza tra a e c come 7"

Di conseguenza cosa vuol dire rilassare un arco?

Dato un grafo $G = (V, E)$, $w : E \to \mathbb{R}$ e a, b, c.

Se pensiamo che per andare da a a c il costo è al più $d[c]$.

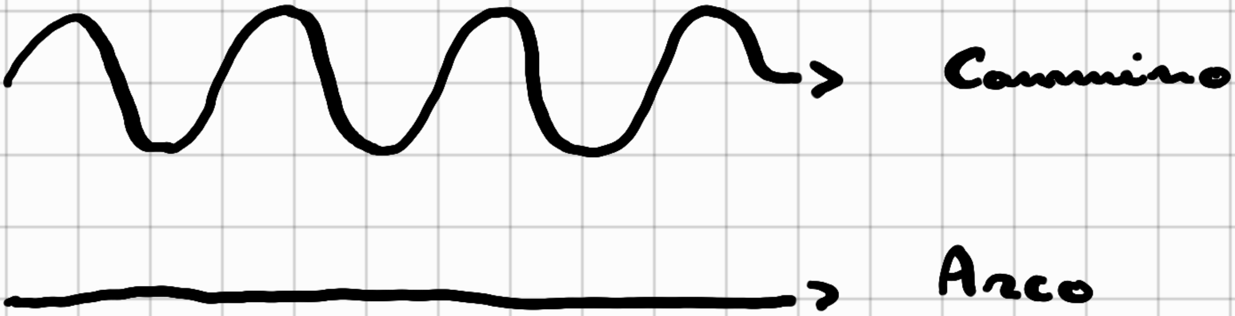
Se pensiamo che per andare da a a b il costo è al più $d[b]$.

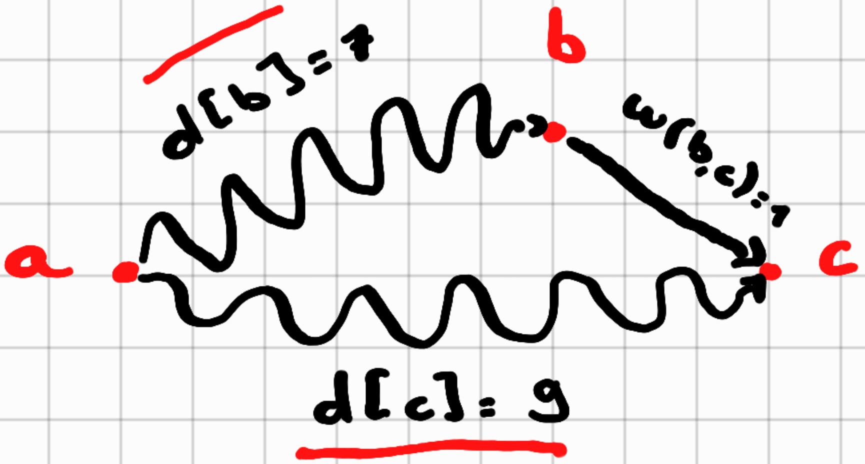
Se l'arco tra b e c ha peso $w(b,c)$.

Se $\underbrace{d[b] + w(b,c)}_{\text{Andare da a in c passando da b.}} < \underbrace{d[c]}_{\text{Quello che pensiamo costi andare da a in c.}}$.

Di conseguenza conviene stimare $d[c]$ come $d[b] + w(b,c)$!

Esempio

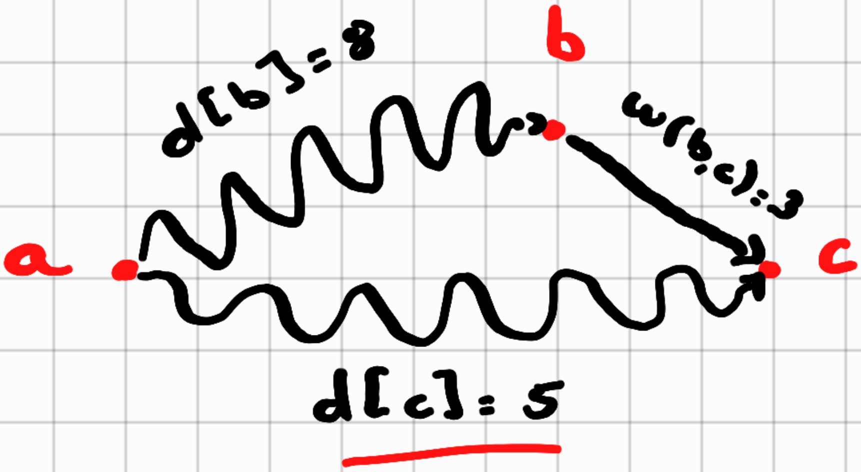




7 + 1 = 8

9 = 9

↳ 8 = 7+1



8+3=11

if $d[c] > d[b] + w(b,c)$

then $d[c] = d[b] + w(b,c)$

Esercizio (sorgente a)

$V = \{a, b, c, d, e\}$ $\quad w: E \to \mathbb{R}_0^+$

$E = \{(a,b,7), (b,c,2), (c,e,4), (e,d,0), (e,b,0)\}$

≥ 0

$\pi = \langle v_1, v_2, v_3, \dots, v_k \rangle$

minimo

$\sigma : \langle v_1, v_2, v_3 \rangle$

(v_2, v_3, v_4)

Tempo 0

sorgente

Q	a	b	c	d	e
d	0	∞	∞	∞	∞
p	NIL	NIL	NIL	NIL	NIL

S $\emptyset$

δ

p

Tempo 1

Q	b	c	d	e
d	7	∞	∞	∞
p	a	NIL	NIL	NIL

$\rightarrow 0 + \underset{w(a,b)}{7} < \infty$

S a

δ 0

p NIL

Tempo 2

Q	c	d	e
d	9	∞	∞
p	b	NIL	NIL

$$7 + \overset{w(b,c)}{2} < \infty$$

S	a	b
δ	0	7
p	NIL	a

Tempo 3

Q	e	d
d	13	∞
p	c	NIL

$$9+4 < \infty$$

S	a	b	c
δ	0	7	9
p	NIL	a	b

Tempo 4

Q	d
d	13
p	e

$$13+0 < \infty$$

S	a	b	c	e
δ	0	7	9	13
p	NIL	a	b	c

Tempo 5

Q $\emptyset$

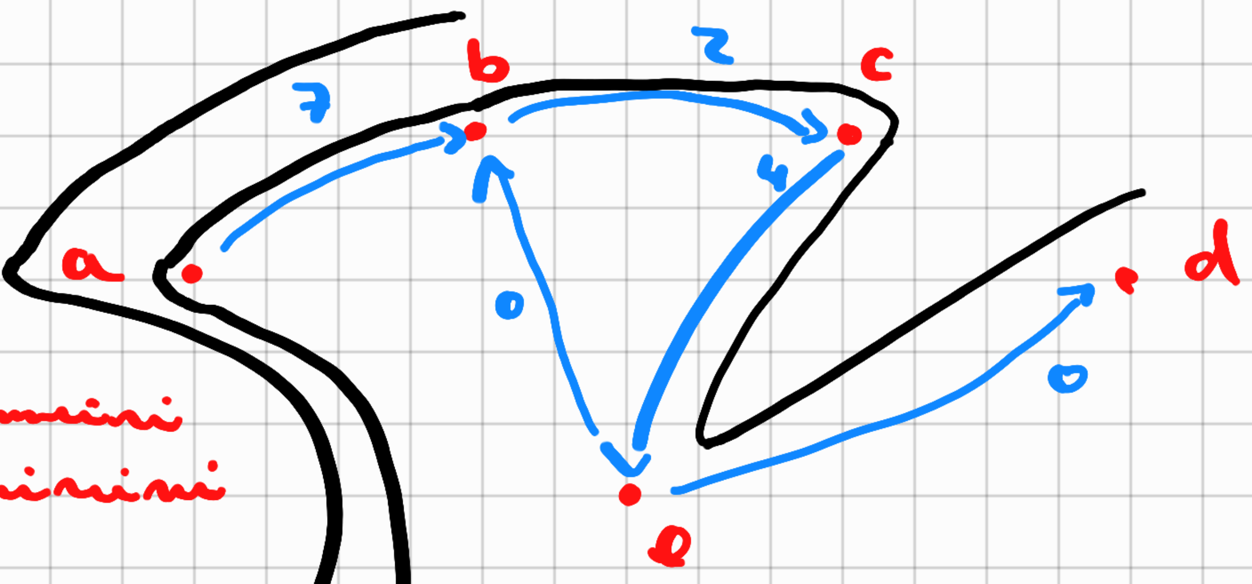
d

p

S	a	b	c	e	d
δ	0	7	9	13	13
p	nil	a	b	c	e

$V = \{a, b, c, d, e\}$

$E = \{(a,b,7), (b,c,2), (c,e,4), (e,d,0),$
$(e,b,0)\}$



Cammini minimi

da	a	
a	b	
a	d	$\longrightarrow \langle a,b,c,e,d \rangle$

SSSP

- **Bellman Ford**

```
for i := 1 up to |V|-1
    ∀ nodo v
        ∀ nodo u
            relax(u,v)
```

- **Dijkstra** noi rilassiamo secondo una coda con priorità

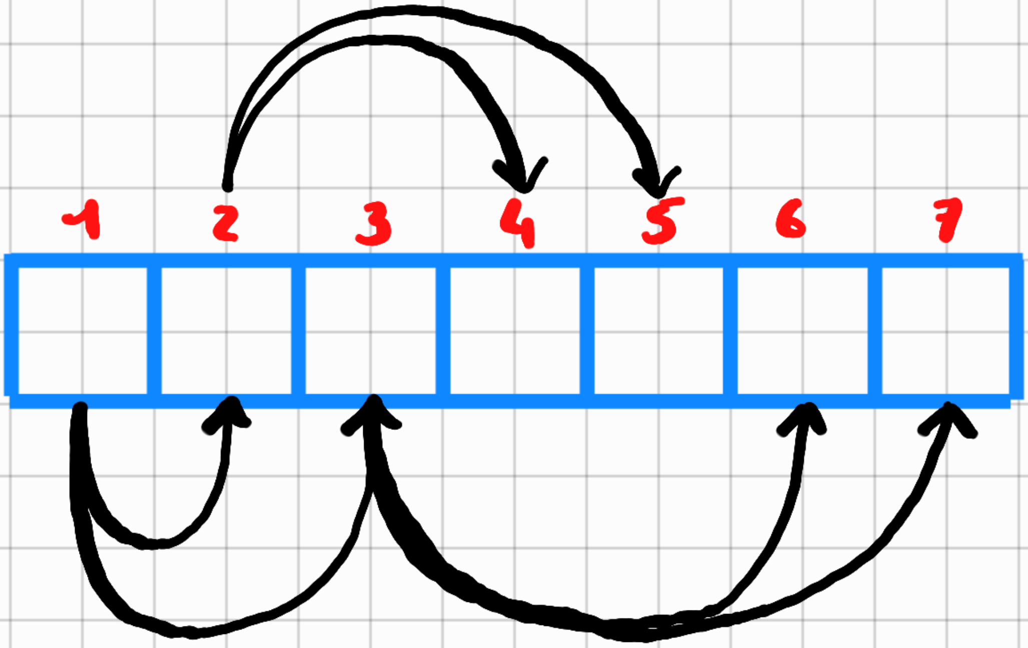
- **Dag** noi rilassiamo in base a un ordinamento topologico

$V = \{a, b, c, d, e\}$

$E = \{(a, b, 7), (b, c, 2), (c, e, 4), (e, d, 0), (e, b, 0)\}$

	T_0	T_1	T_2	T_3	T_4
d[a]	0	0	0	0	0
d[b]	∞	7	7	7	7
d[c]	∞	∞	9	9	9
d[d]	∞	∞	∞	∞	13
d[e]	∞	∞	∞	13	13

Heap



1 Insert 3

2 Insert 8

3 Insert 9

4 Insert 4

5 Insert 5

6 Extract Min

7 Insert 2

8 Extract min

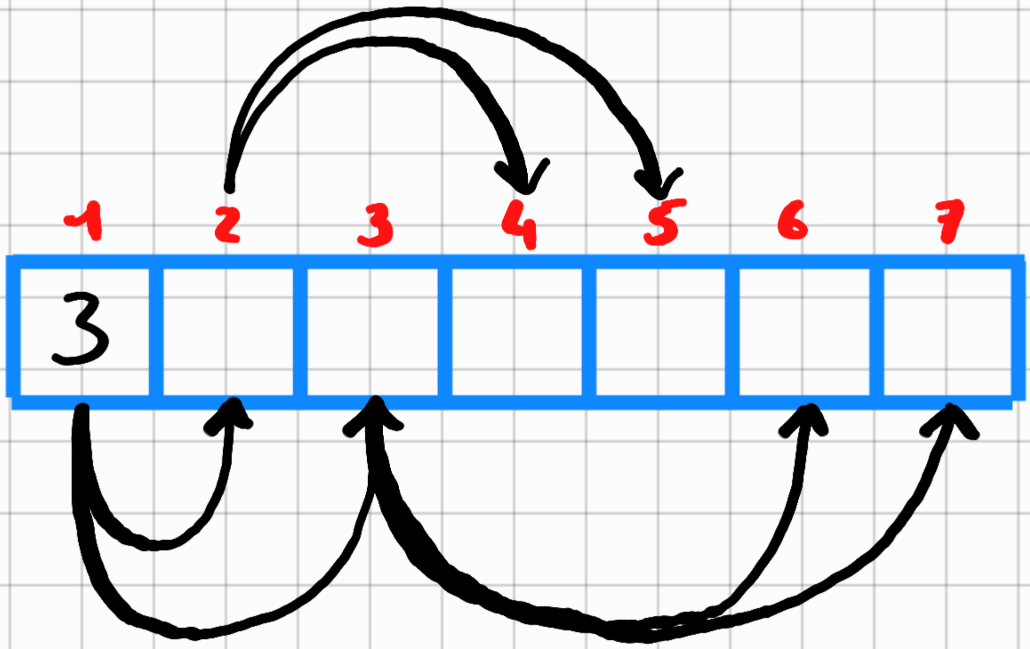
9 Extract min

10 Extract min

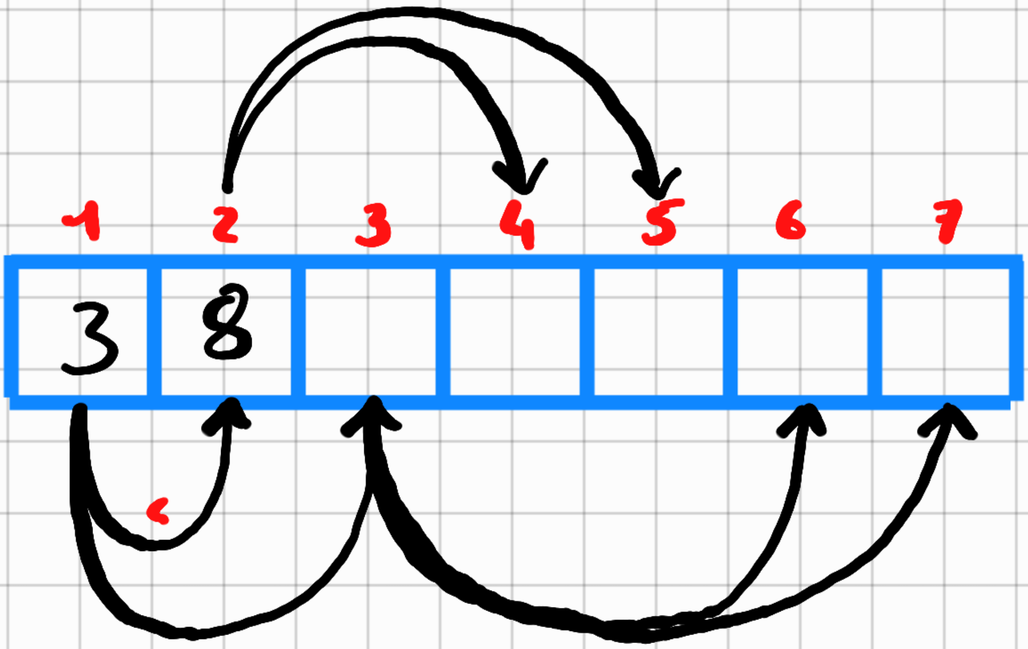
11 Extract min

12 Extract min

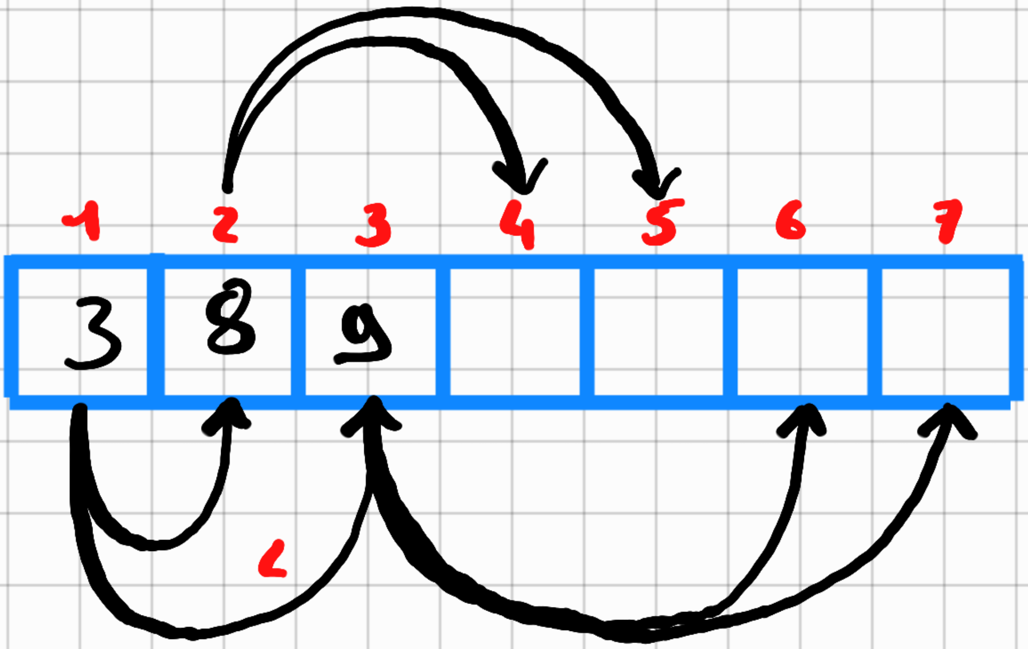
Tempo 1



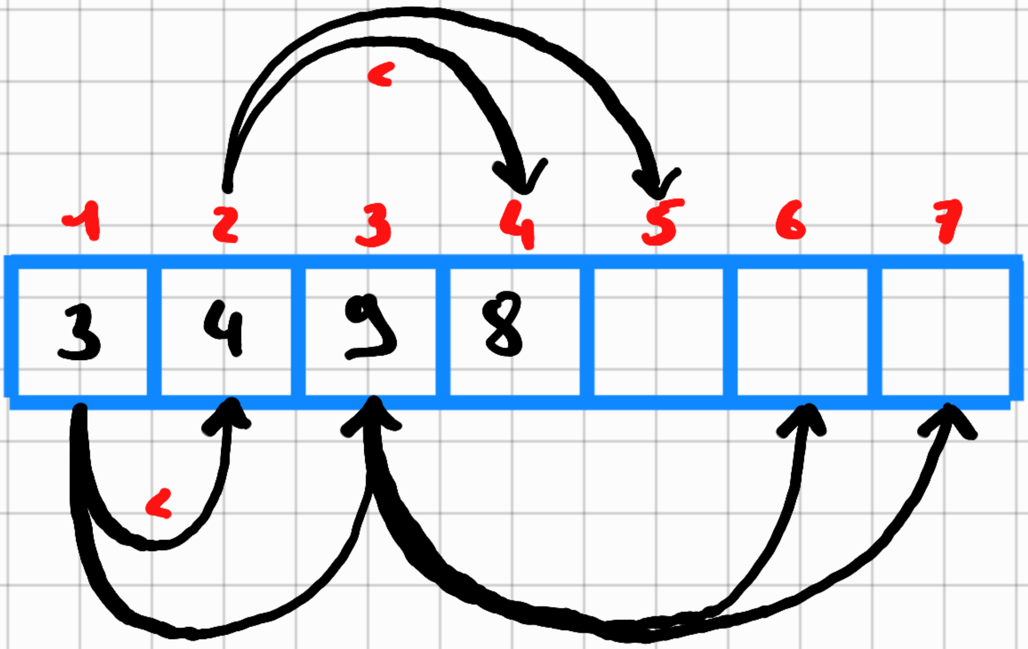
Tempo 2



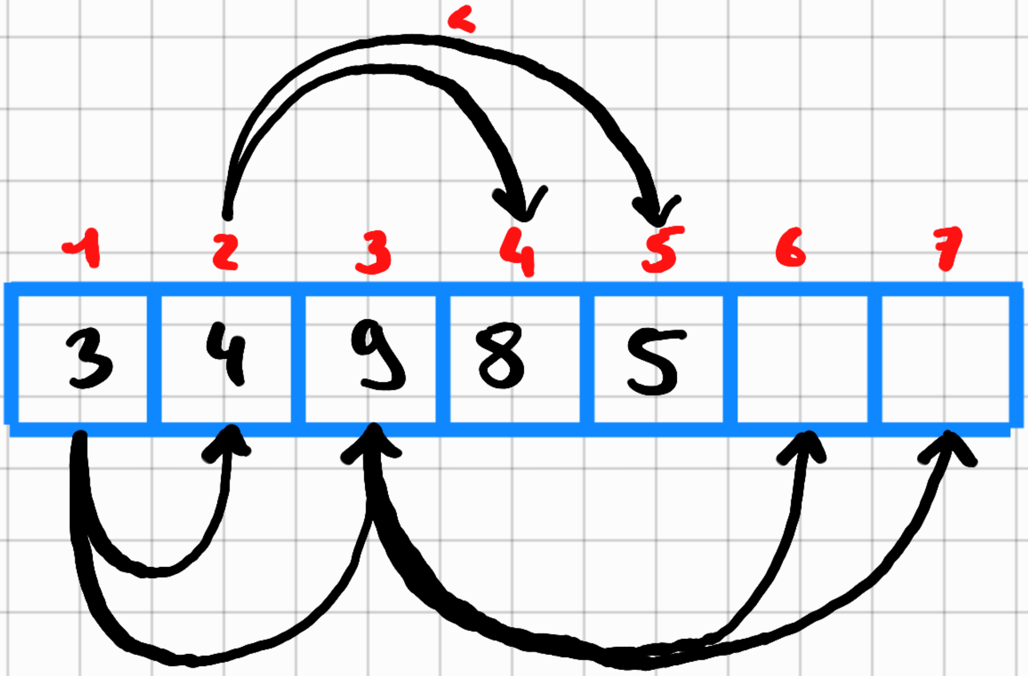
Tempo 3



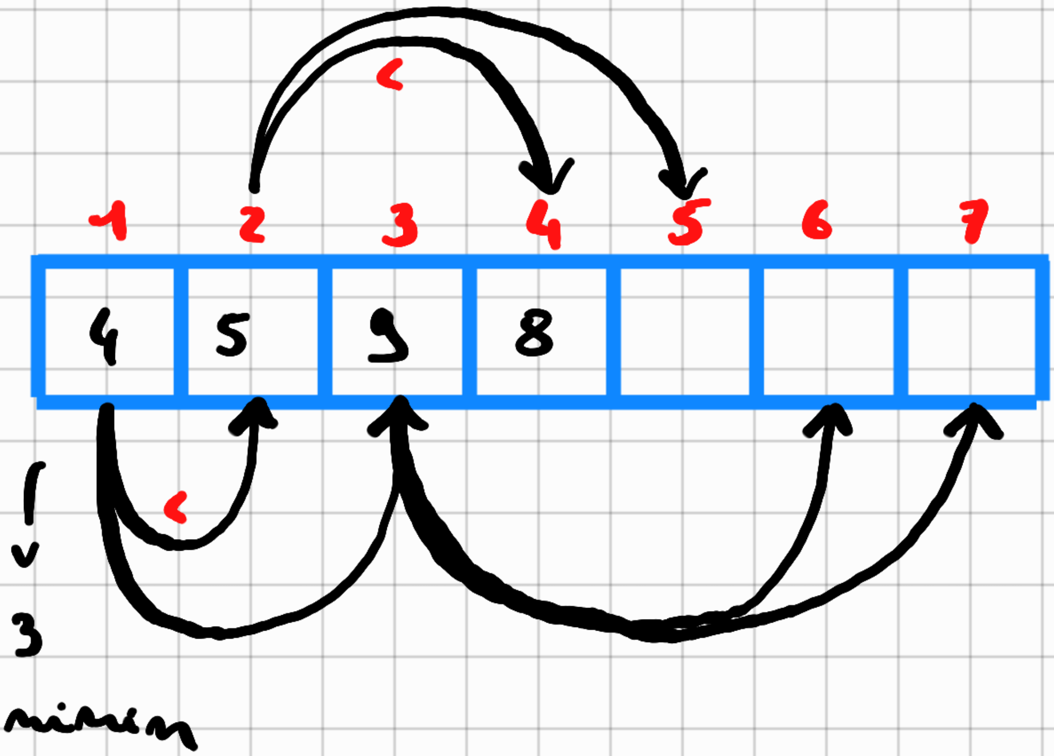
Tempo 4



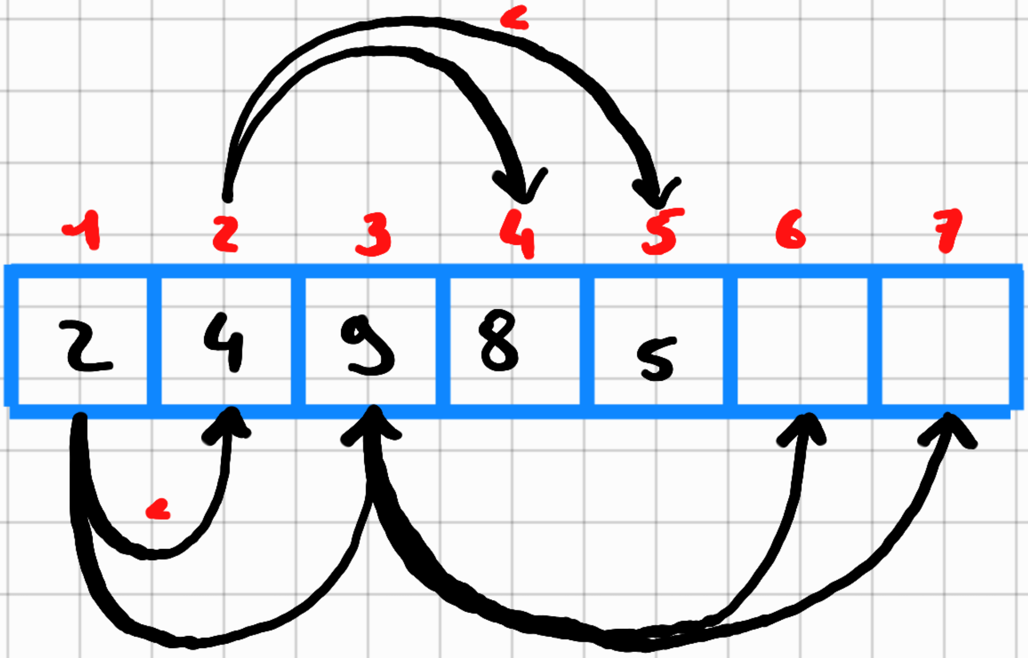
Tempo 5



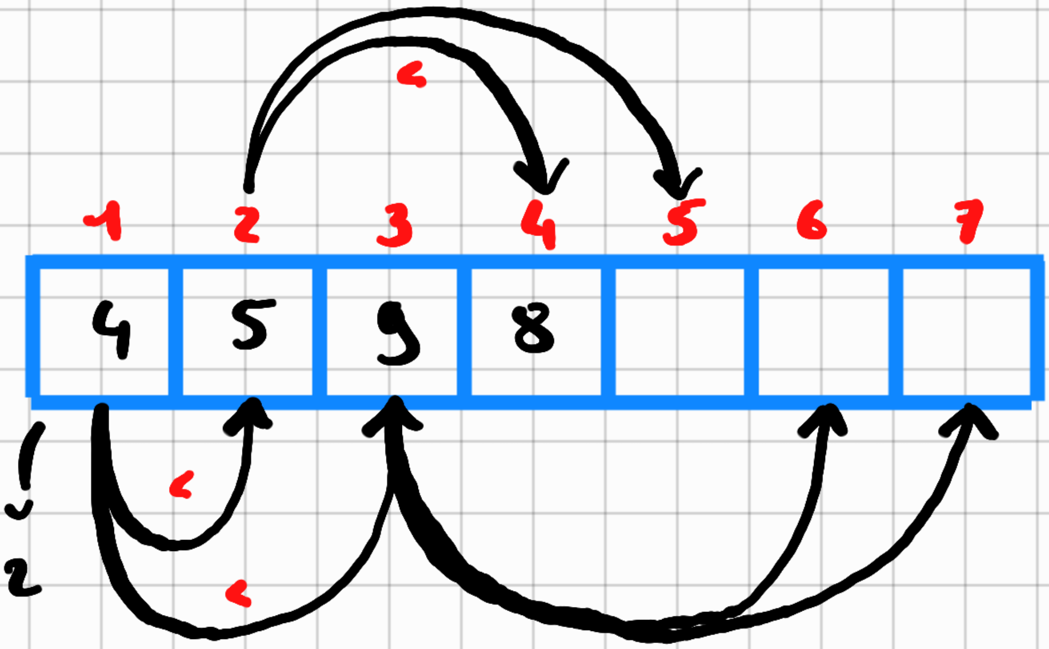
Tempo 6



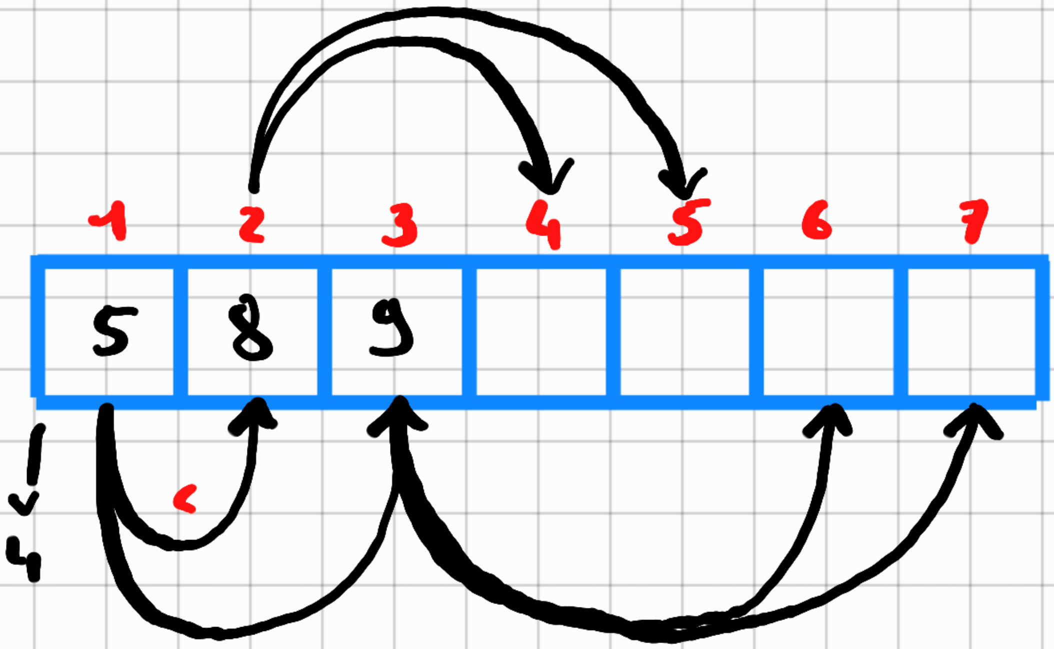
Tempo 7



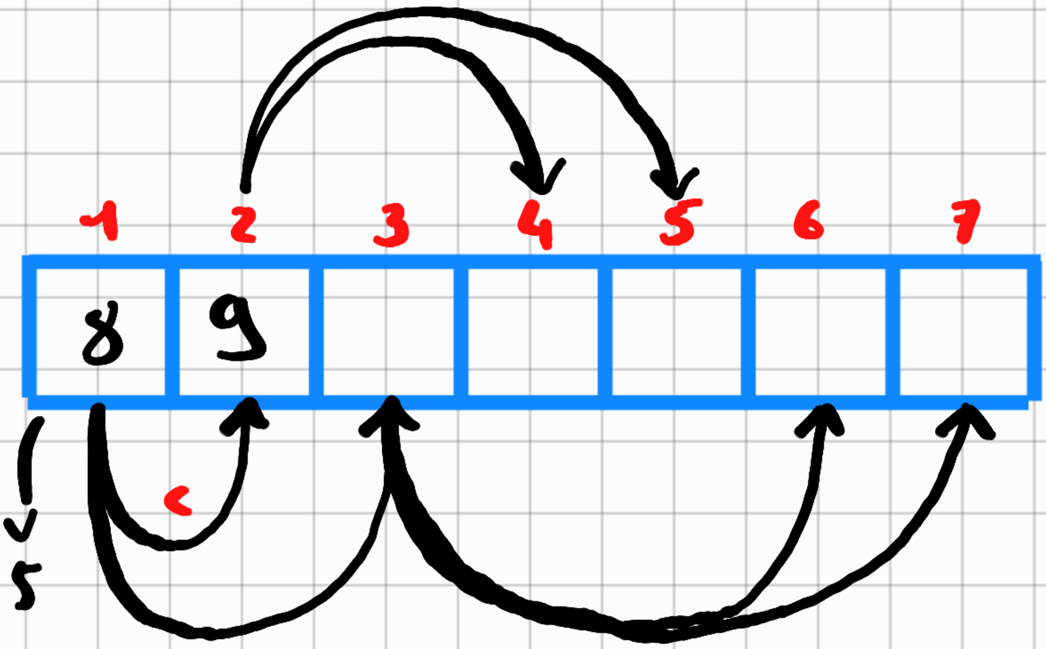
Tempo 8



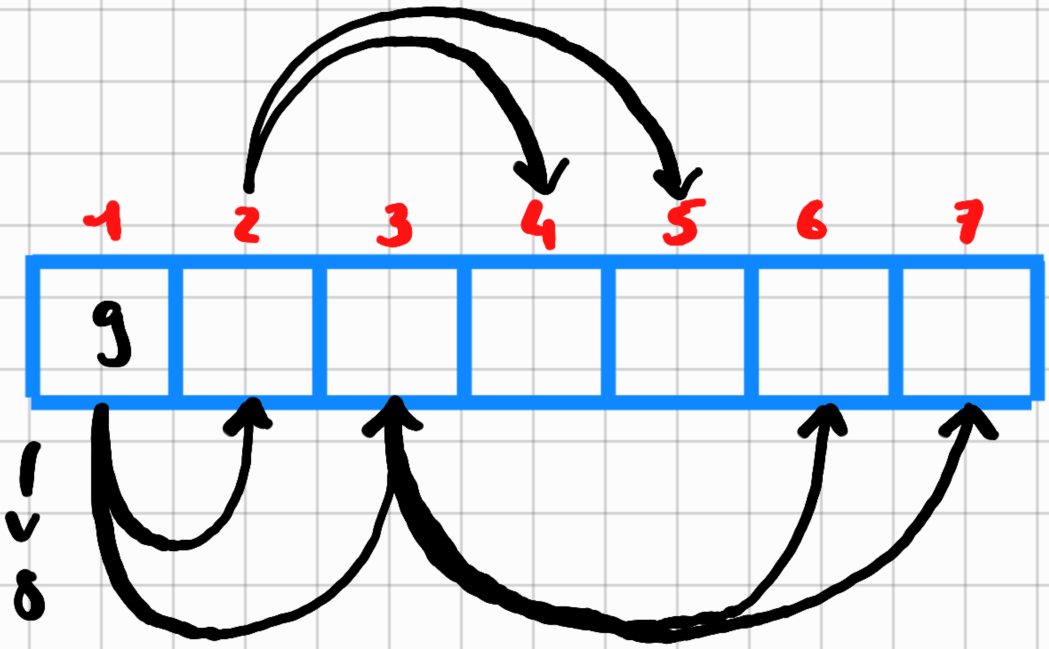
Tempo 9



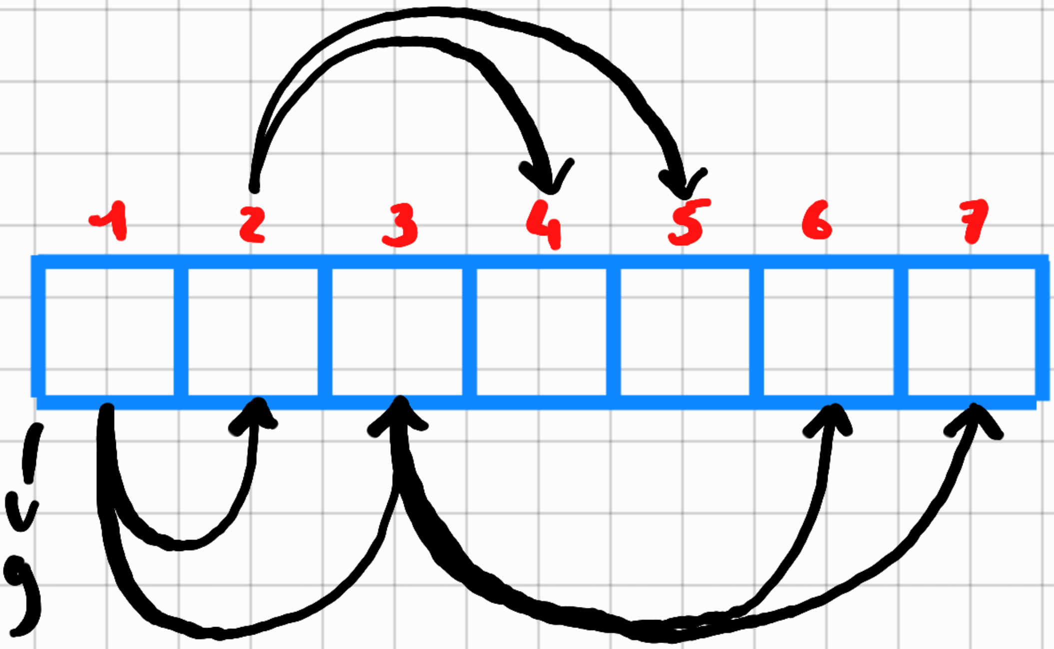
Tempo 10



Tempo 11

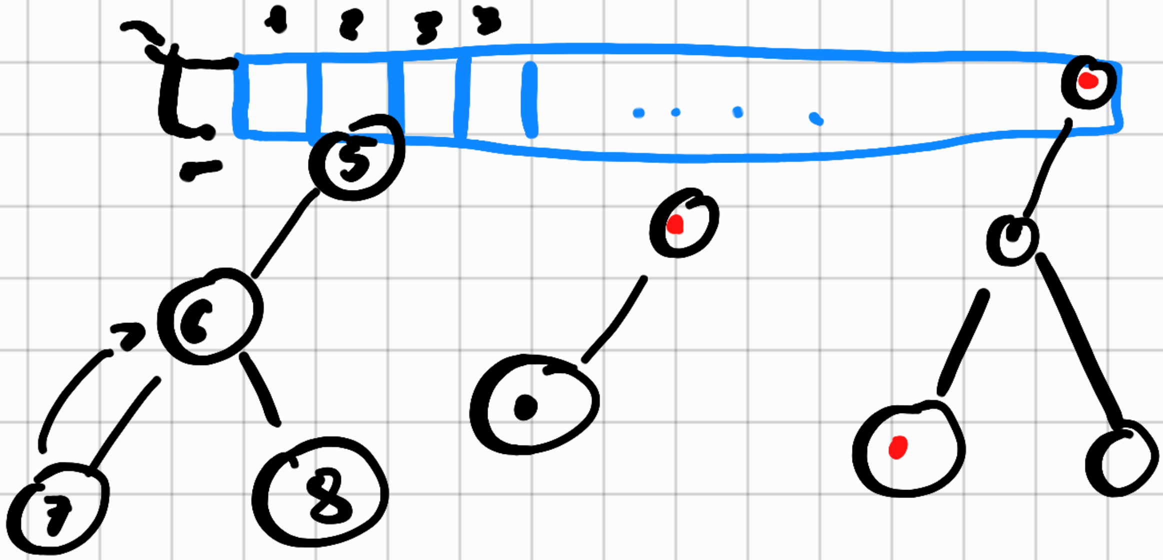


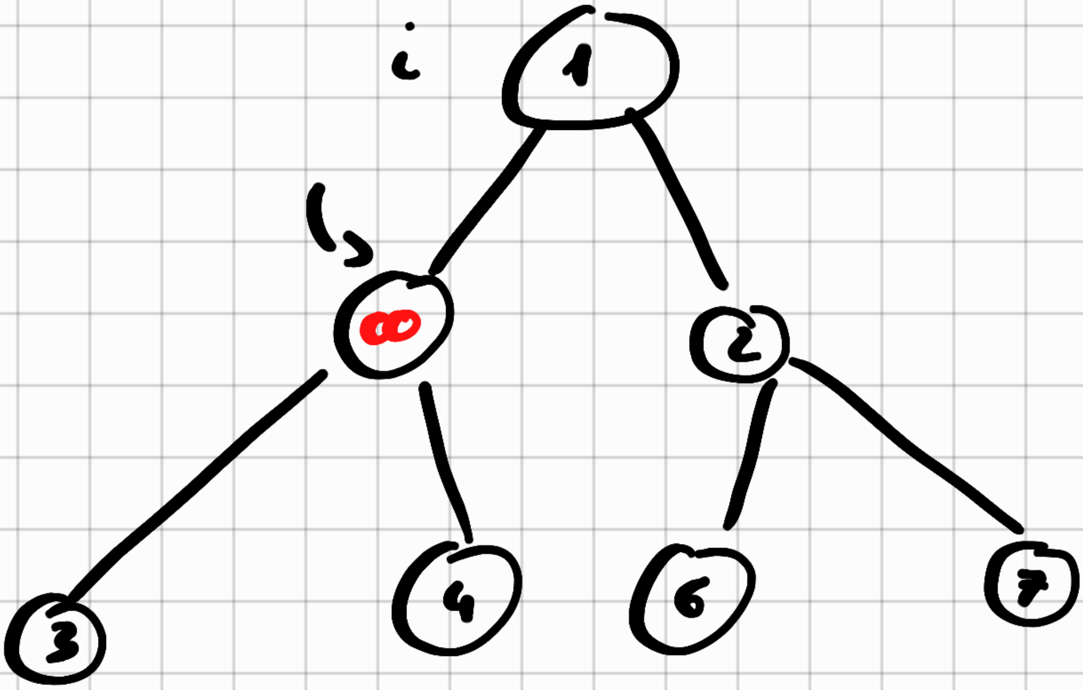
Tempo 12



Implementiamo...

Heapify




i
1
∞
2
3
4
6
7

Esercizi

Prima su un max heap e poi su un min heap:

1 Insert 10
2 Insert 20
3 Insert 7
4 Insert 8
5 Insert 9
6 Insert 6
7 Extract
8 Extract
9 Insert 1
10 Extract 1

Applicare Bellman-Ford e Dag

$V = \{a, b, c, d, e\}$ sorgente e

$E = \{ (a, c, 7), (a, b -2), (d, e, 7), (e, a, -1), (e, b, 2), (b, c, -1) \}$

Applicare Bellman Ford e Dijkstra

$V = \{a, b, c, d, e\}$ sorgente e

$E = \{ (a, c, 7), (a, b\ 2), (d, e, 7), (e, a, 1), (e, b, 2), (b, c, 1) \}$